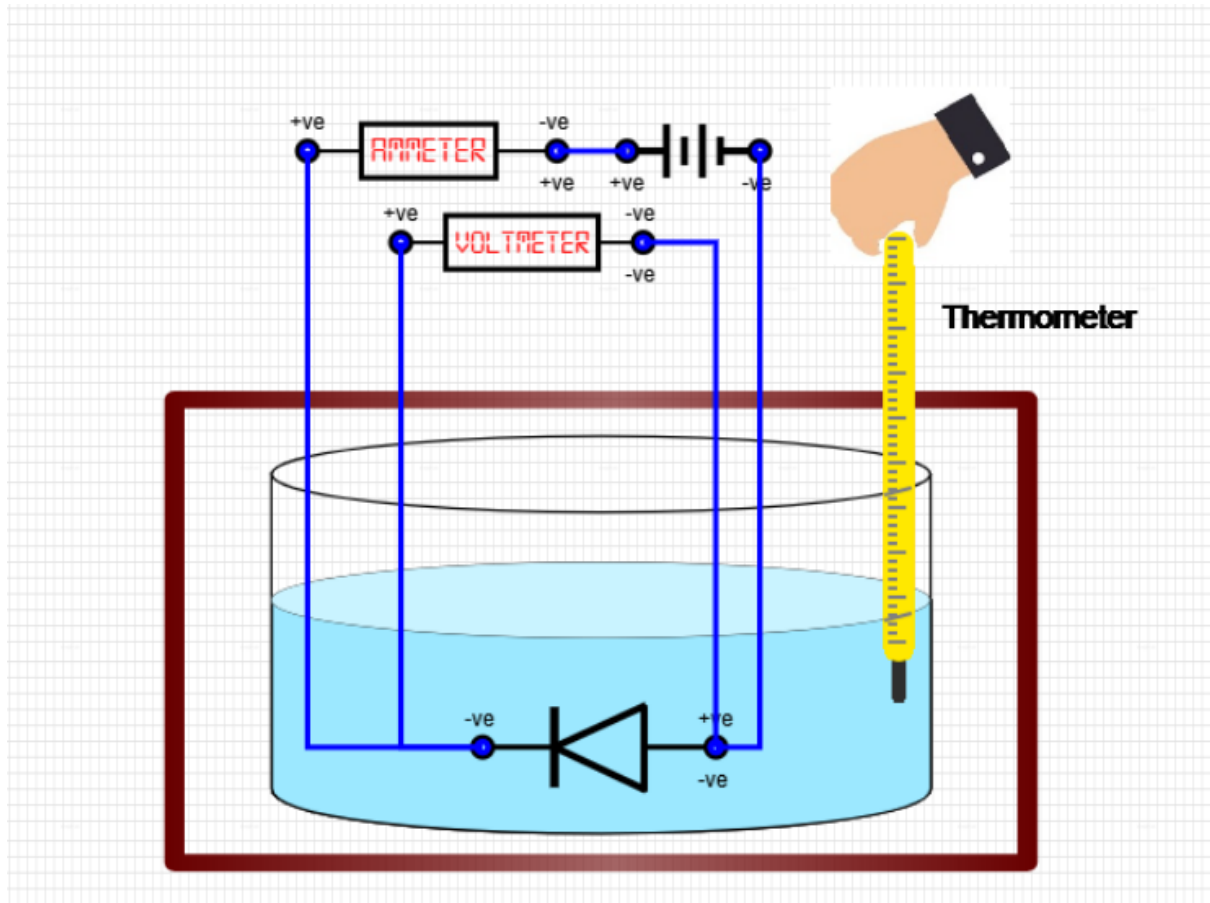
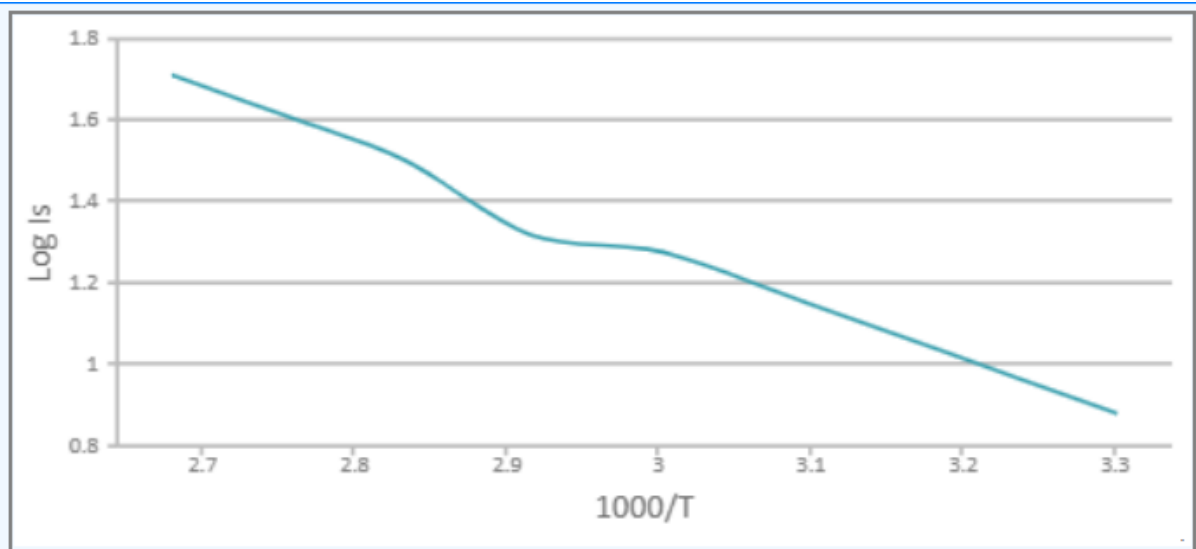




Tabular Column

S.No.	Temperature(°C)	Current I_s (μ A)	Temperature(°K)	$10^3/T$	$\log_{10} I_s$
01	30	7.568	303	3.30	0.879
02	40	10.476	313	3.19	1.020
03	50	14.214	323	3.10	1.153
04	60	18.934	333	3.00	1.277
05	70	20.805	343	2.92	1.318
06	80	32.001	353	2.83	1.505
07	90	40.711	363	2.75	1.610
08	100	51.126	373	2.68	1.709

Graph:



Calculated Slope is **-0.743**

NOTE : Enter value of Slope calculated above in the below box

Energy Band Gap (E_g) = / 5.04

Observed Energy Band Gap is **0.147**

$$E_g = 2 \times 2.303 \times 0.147$$

$$= 0.677 \text{ eV}$$

$$E_g = 0.677 \text{ eV}$$

Result : Energy band gap of the given semiconductor by simulation method is found to be = 0.677eV

The given semiconductor is germanium(Ge).